MINSPIRE 2026 CAPSTONE

Climate zones and forecasting in Nepal

What 39 years of daily data can and can't tell us

THE PROBLEM

Three questions

1

Can data alone group Nepal's districts into climate zones?

2

How well can we forecast monthly temperature and rainfall, and is it harder in some zones?

3

How far can we trust the data we are using?

THE DATA

62 districts, 39 years, one big caveat

62

districts

39

years, 1981 to 2019

883,128

daily rows

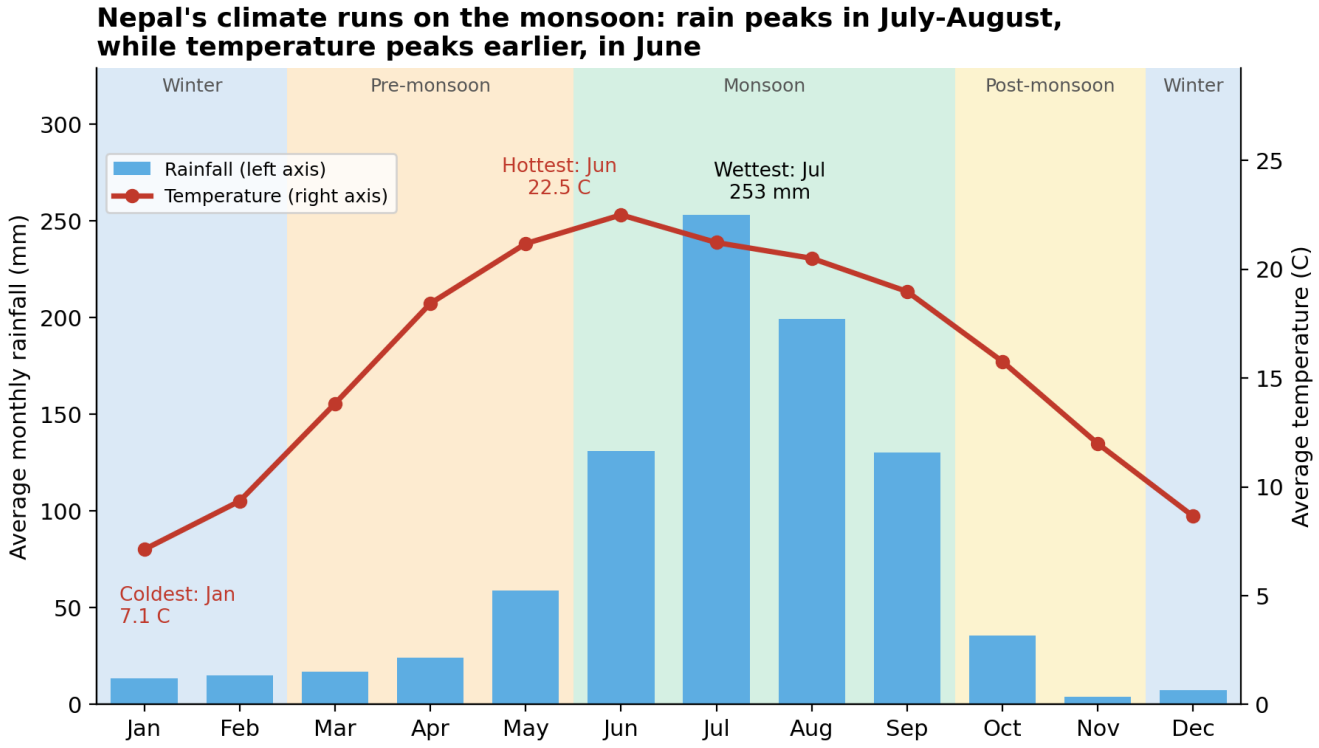
0

missing values

Source: NASA POWER via Kaggle. This is **modelled data**, not station readings, so changes in how it was produced can create artificial jumps.

EXPLORING THE DATA

The monsoon runs the year

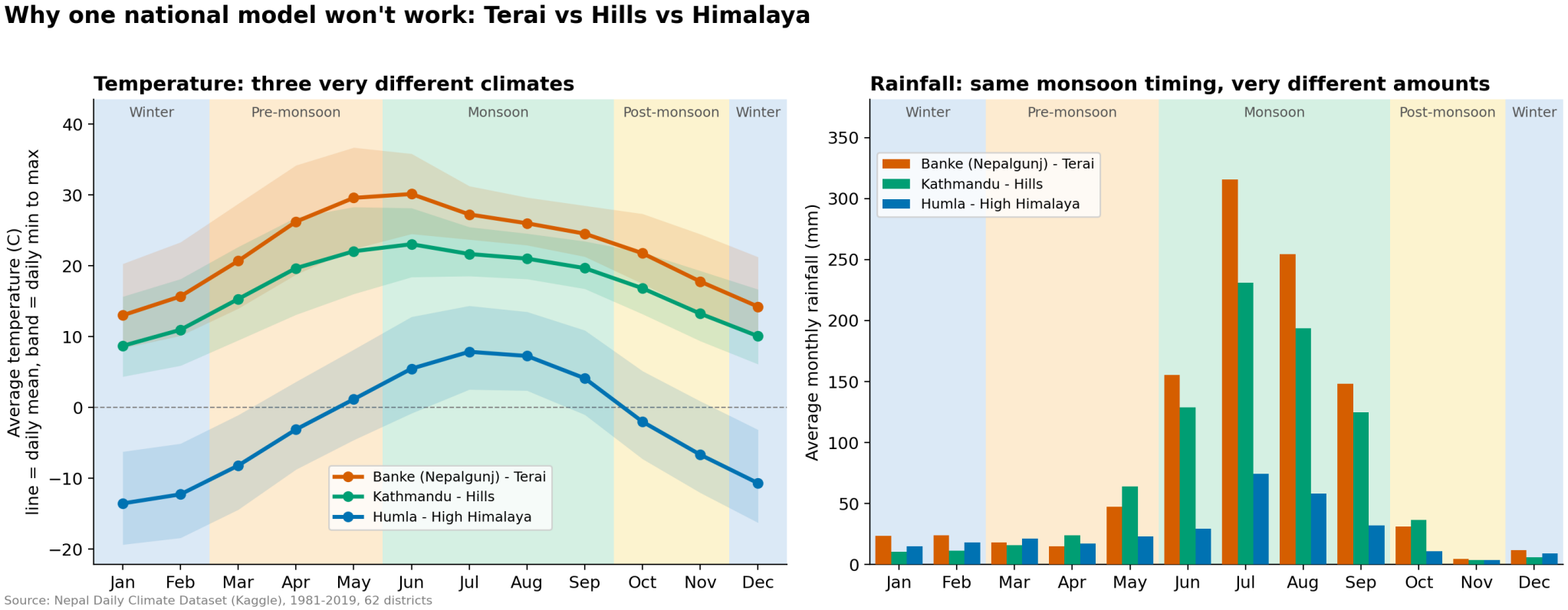


Source: Nepal Daily Climate Dataset (Kaggle), 1981-2019, 62 districts

About 80% of rain falls June to September; July is the wettest month, June the hottest.

EXPLORING THE DATA

Nepal's regions are very different climates

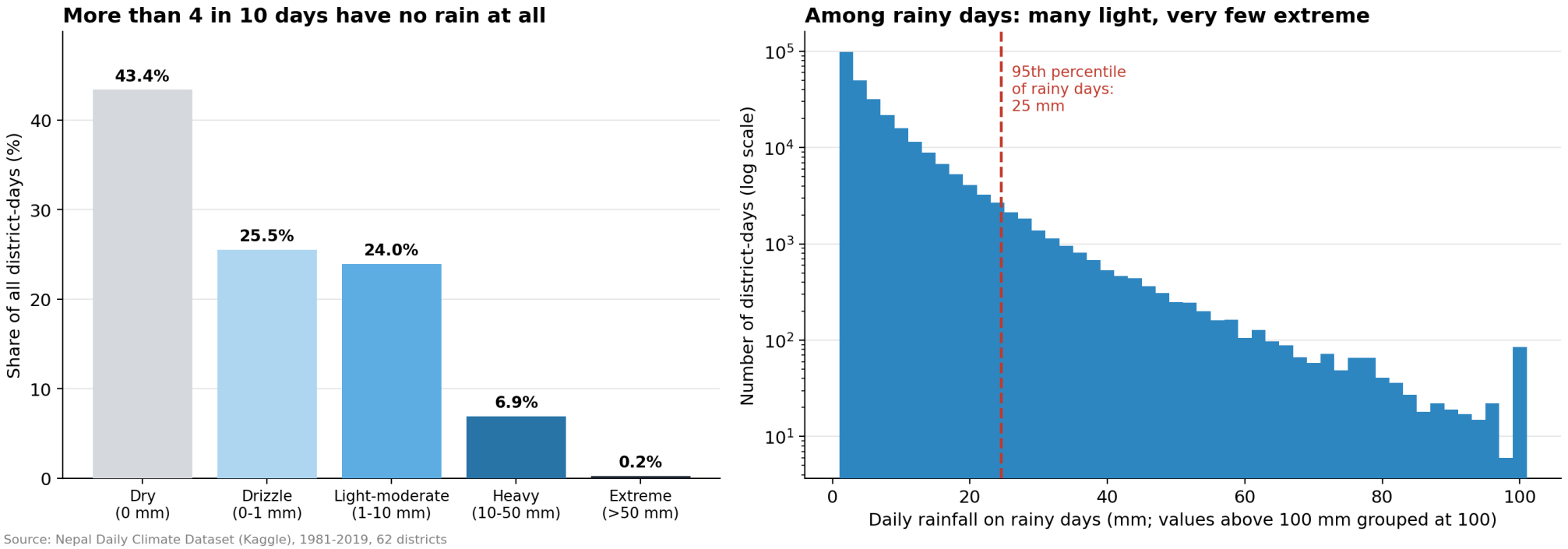


Banke (Terai) about 22 °C, Kathmandu (hills) about 17 °C, Humla (Himalaya) below 0 °C. One national model would not work.

EXPLORING THE DATA

Rainfall is spiky: 43% of days are dry

Rainfall is highly skewed, which makes it hard to forecast



Source: Nepal Daily Climate Dataset (Kaggle), 1981-2019, 62 districts

Most rainy days are light and a few exceed 50 mm. There is no significant warming trend (+0.014 °C per decade, p = 0.80).

METHOD

Find zones, forecast, then stress-test

STEP 1
Daily data
62 districts

›

STEP 2
Four features
per district

›

STEP 3
K-Means
3 climate zones

›

STEP 4
Forecast 2015 to 2019
baselines vs SARIMA

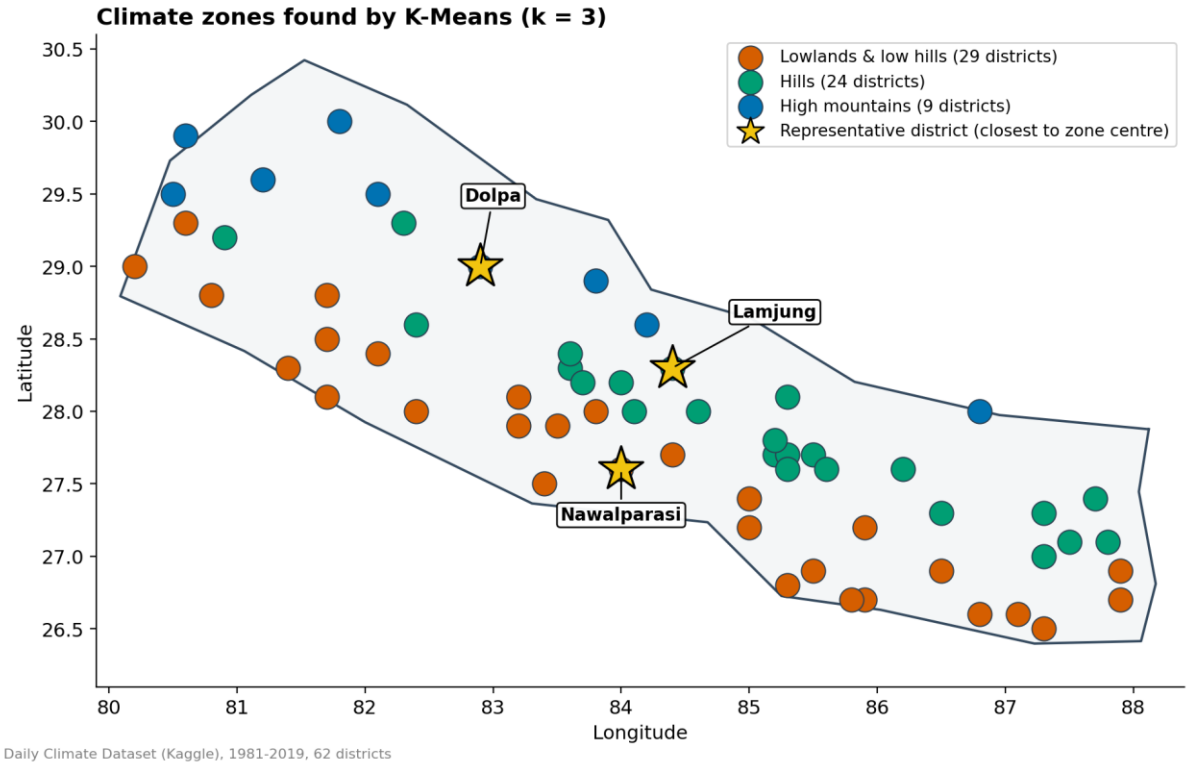
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STEP 5
Stress tests
districts, windows, methods

Features: mean temperature, annual rainfall, monsoon share, seasonal temperature swing. Models train only on earlier years and are tested on 2015 to 2019.

RESULTS: ZONES

The data finds three climate zones



Source: Nepal Daily Climate Dataset (Kaggle), 1981-2019, 62 districts

29 districts

lowlands and low hills

24 districts

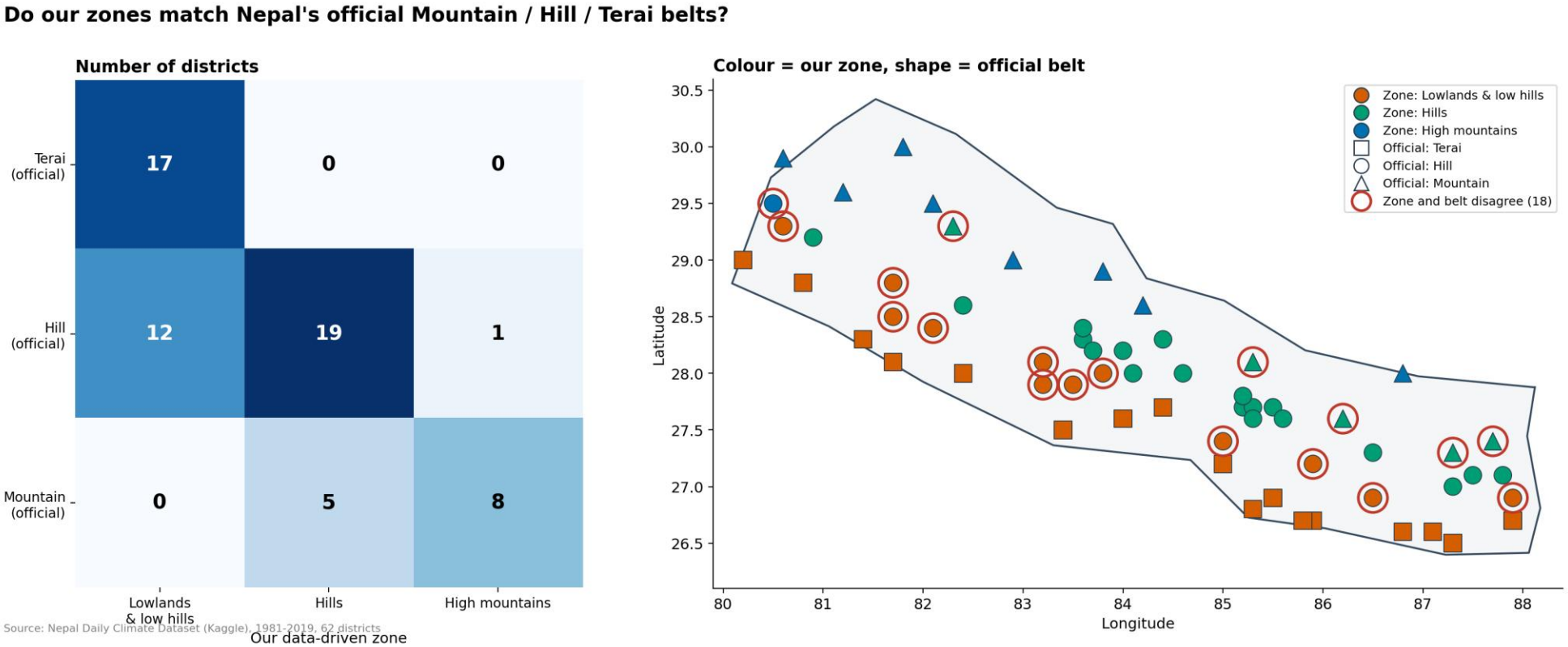
hills

9 districts

high mountains

RESULTS: ZONES

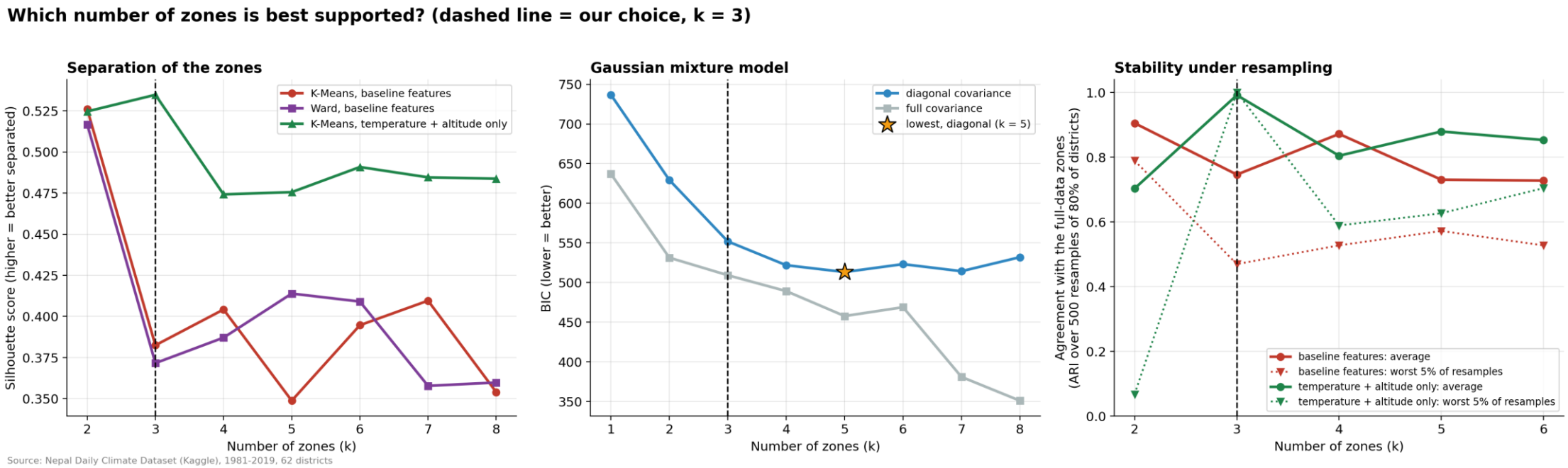
The zones follow the Terai, Hill and Mountain belts



All 17 Terai districts land in the lowland zone and 71% of districts match overall. Disagreements sit at the hill/mountain edge.

RESULTS: ZONES

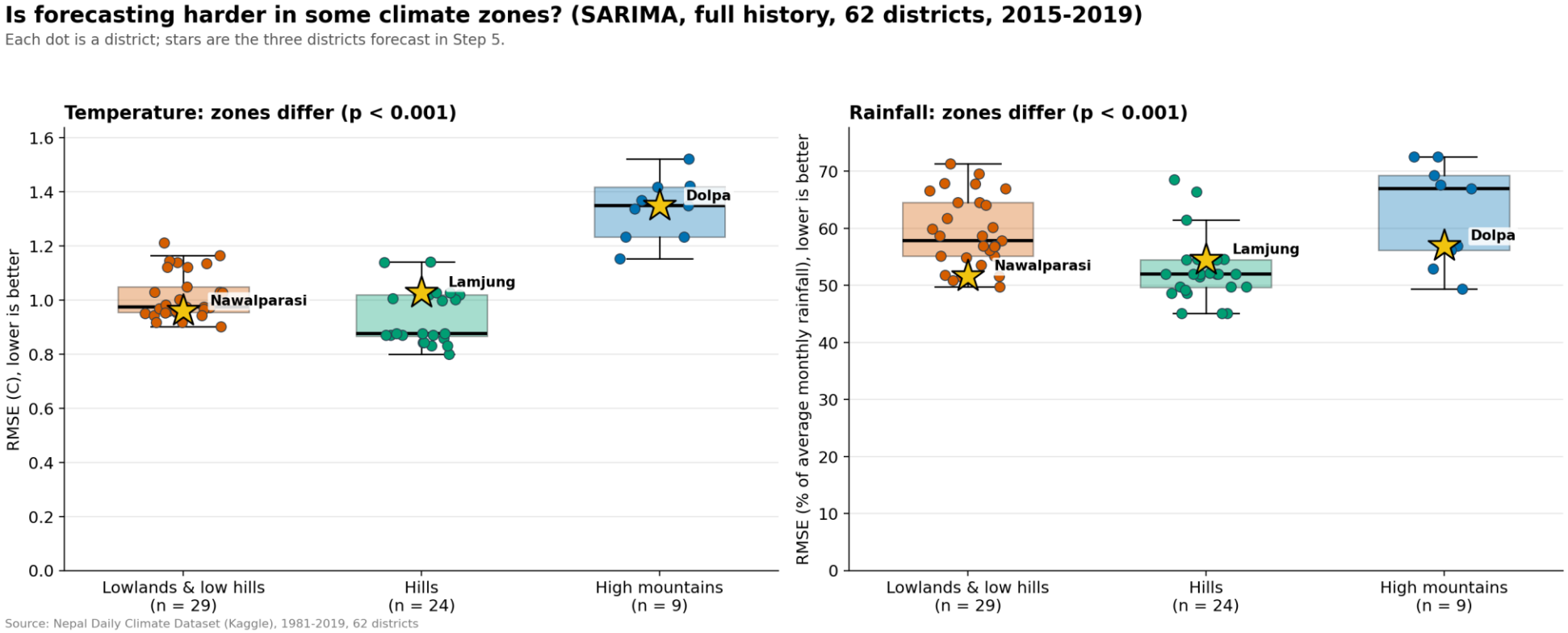
Temperature and altitude support exactly three zones



With temperature and altitude only, k = 3 is the clear winner (silhouette 0.54, stability 0.99). With rainfall features, k = 2 scores higher.

RESULTS: FORECASTING

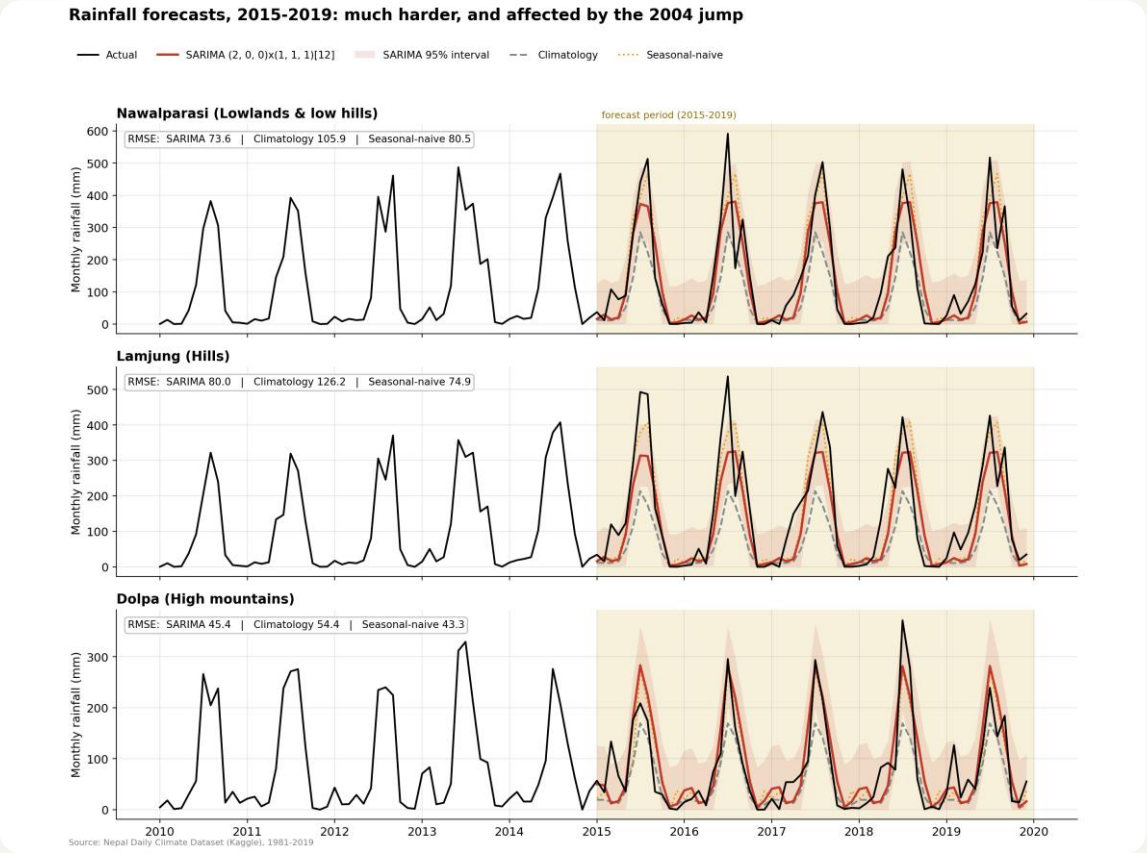
Forecasting is easiest in the hills



SARIMA error for all 62 districts, 2015 to 2019. Temperature error is about 1 °C; rainfall error is about 50 to 65% of average monthly rainfall.

RESULTS: FORECASTING

Rainfall is much harder than temperature



SARIMA beats the baselines

in most districts

Temperature

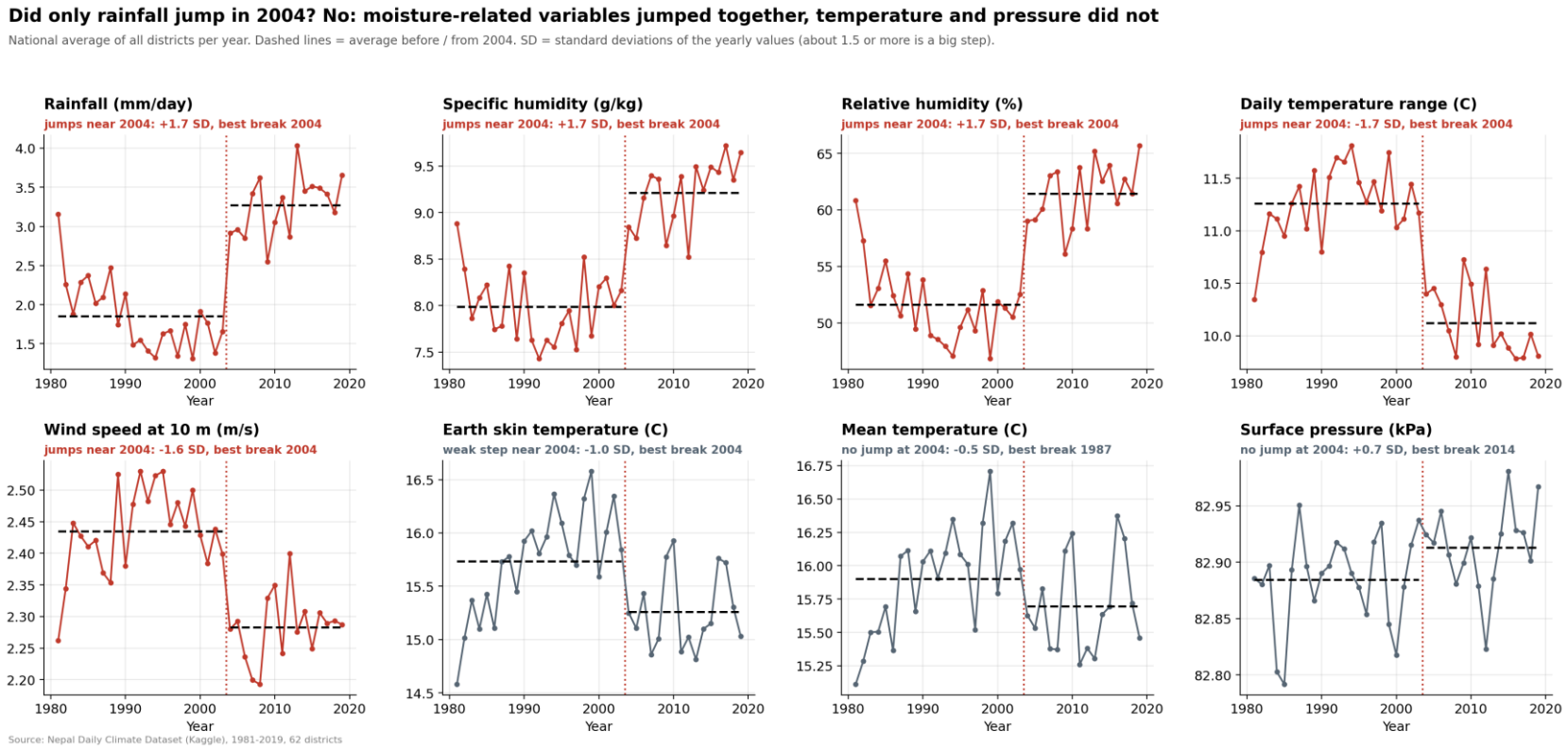
mostly the seasonal cycle

Full-history monthly average

runs about 30% too low

FAILURE ANALYSIS

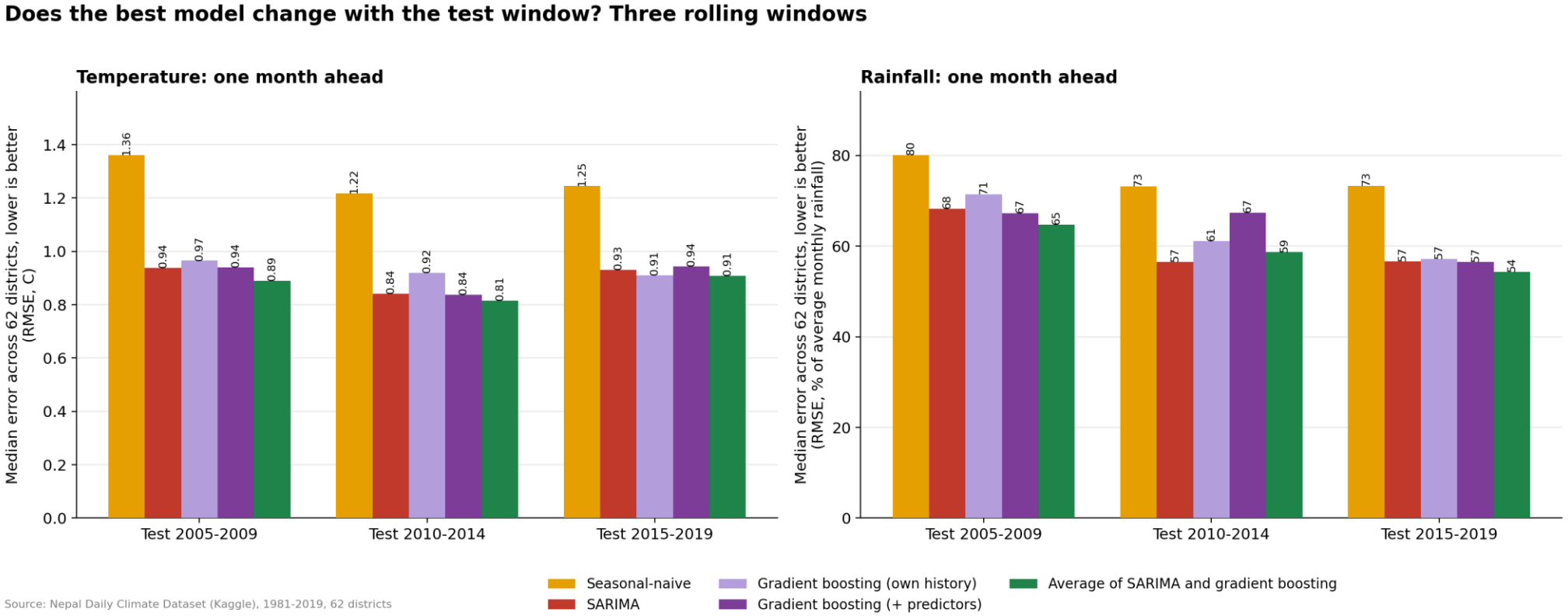
A hidden jump in 2004



Seven moisture, wind and temperature-range variables jump together (Dec 2003 to May 2004), mean temperature and pressure do not. Most likely a data artifact, unproven.

FAILURE ANALYSIS

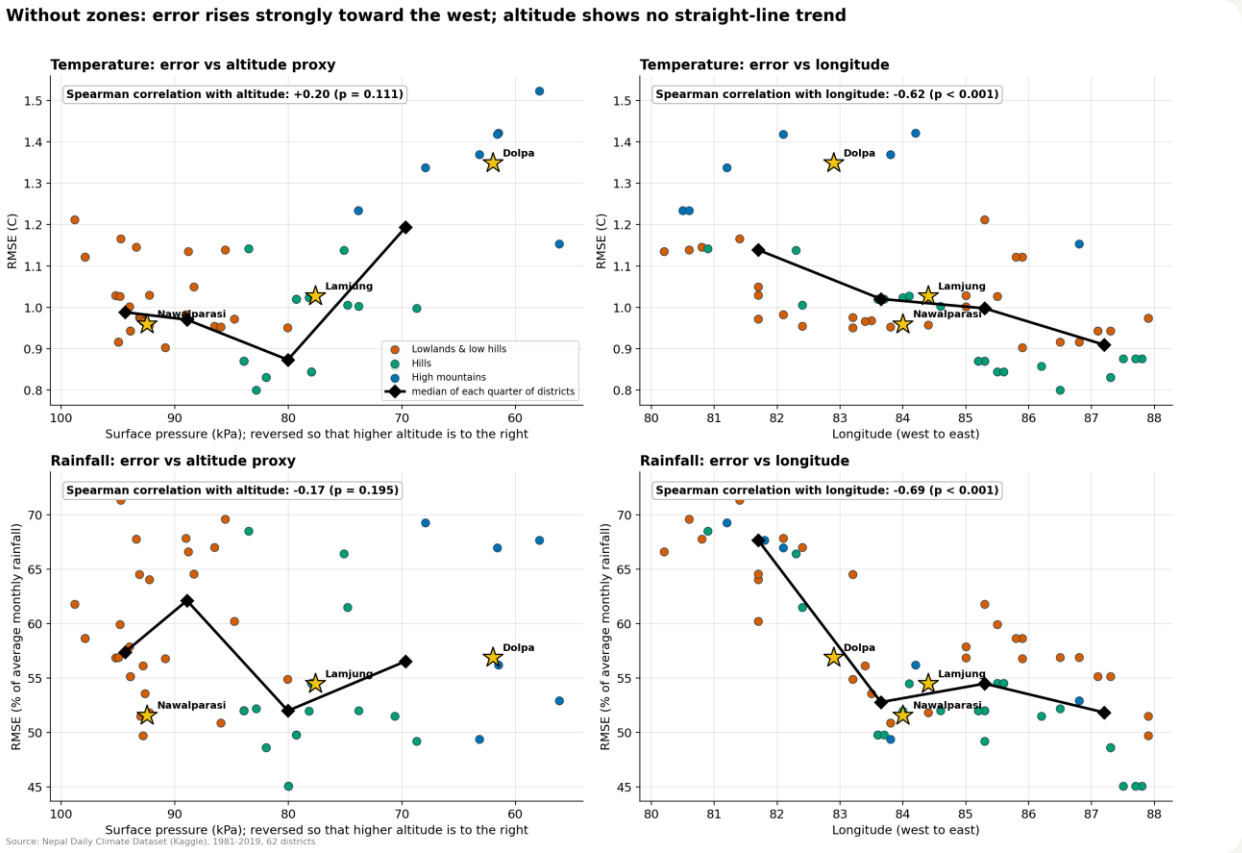
Fancier models gave only small gains



Averaging SARIMA and gradient boosting was best in 5 of 6 tests, by about 2 to 5%. A log transform made rainfall worse.

FAILURE ANALYSIS

Forecast error rises toward the west



-0.62 and -0.69

correlation with longitude (temperature, rainfall)

No straight-line altitude trend

temperature is U-shaped: hills easiest

Cause untested

weaker monsoon? modelled data?

WHAT NEXT

Limitations and next steps

Limitations

- Modelled data with a 2004 step
- Mountain zone has only 7 to 9 districts
- One main test window, 2015 to 2019
- West-to-east cause not tested

Next steps

- Check 2004 against station records
- Add elevation data
- Separate western and eastern models
- Test longer forecast horizons

CONCLUSIONS

1. Three climate zones along temperature and altitude
2. Temperature is easy to forecast, rainfall is hard, and the west is harder
3. Checking the data mattered as much as building models

Thank you. Questions?